A Machine-Actionable Rights Expression Vocabulary for Open Science

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Version 0.5.2

OpenREL turns the rights and rules that govern research objects from prose that only people can read into structured statements that systems can interpret and act on. This first public draft presents its architecture, policy model, vocabulary and Knowledge Base, with worked examples, to collect community feedback and support early implementations in the European Open Science Cloud.

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Executive Summary

In open science, the rights that govern a research object are still largely expressed for human readers. A dataset typically carries a licence label, a repository page states access conditions in prose, and a data management plan describes processing obligations in a document. Machines that move data across the European Open Science Cloud cannot reliably act on any of this. They can read a licence identifier, but they cannot tell whether a specific request should be permitted, whether an obligation must be met first, or whether two policies attached to the same object are in conflict.

OpenREL addresses this gap. It is an open, machine-actionable rights and policy framework that lets the full range of conditions governing access to and reuse of a digital resource be expressed, exchanged and evaluated in a single structured representation. Licence terms, access conditions, the requirements of data subjects, ethical and legal constraints, and the operational rules of the platforms that disseminate a resource can all be stated together, rather than as free text managed outside the licence. It is built on the W3C ODRL 2.2 standard and expressed in RDF-TTL and JSON-LD, and it returns a clear outcome for policy comparisons and combinations of PERMIT, CONDITIONAL or DENY, always with a structured reason. Policy collections may reference constituent policies and preserve comparison and negotiation records, including unsuccessful requests.

The specification rests on three pillars. First, an ODRL-based policy model distinguishes licences, access rules, processing rules and contextual policies, supports collections of policies, and provides structures for recording policy-level evaluations and their evidence. The ontology defines how comparisons, assertions, parameter resolution and evaluation results are represented; an evaluator or associated service must perform the actual comparisons and make decisions. Second, a version-controlled vocabulary and Knowledge Base provide reusable actions, constraints, parameters, assertions and alignments with related vocabularies. Third, authoring tools, including the Licence Wizard, support the preparation of machine-readable policies.

Specific extensions to the encoding capabilities offered by ODRL include concept schemes for right operands and its parameterisation linked to specific end user, resource, or platform contexts, cross-vocabulary mappings/crosswalks (including to DALICC and Creative Commons vocabularies), and reusable Named Constraints as explicit design mechanisms.

This draft presents the architecture and the specification so that others can test it and comment on it. Where a capability is planned rather than delivered, the document will indicate this. The intended readers are repository developers, data space architects and standards communities who need a stable reference, together with data stewards, policy professionals and reviewers who need to understand what OpenREL expresses and how far it has progressed.

Unlike existing rights vocabularies, OpenREL is designed not only to describe rights, but to support operational policy evaluation across heterogeneous research infrastructures through a common machine-actionable representation.

Document Status

This document presents the first public draft of the OpenREL Specification. The document body introduces contextual and illustrative information on the OpenREL architecture, policy model, vocabulary and implementation approach. It is published to collect community feedback, support early implementations, and provide a stable reference for discussion. It does not represent the final OpenREL Specification.

The annexed stable OpenREL Specification, version 0.5.2, provides the technical reference: Annex A defines the vocabulary and policy model, Annex B specifies policy evaluation, and Annex C addresses contextual assertions and parameter resolution. The normative Version 1.0 is expected alongside EOSC Beyond Deliverable D14.3 in early 2027.

The draft follows the maturation model used by standards bodies such as the W3C First Public Working Draft. Several components are described here as work in progress: the vocabulary is actively growing, the graph database is under development, and native repository integration remains an open gap. These are stated openly throughout the document so that implementers can distinguish what exists today from what is planned.

Relationship to Prior Work

This specification does not restate the state of the art. The analysis of existing rights expression approaches, including ODRL, ccREL, MPEG-21, RightsStatements and repository rights metadata, together with the gap analysis that motivates OpenREL, is documented in EOSC Beyond Deliverable D13.3 and will be cited by DOI upon publication of that deliverable. The present document focuses on OpenREL itself and refers to D13.3 where the underlying evidence is relevant.

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Glossary

The glossary provides disambiguation for terms that are used interchangeably. Where applicable, terms that are used by preference in this document are shown in bold.

Term	Explanation
Resource, Asset, Work, Creative Work	These terms all refer to the resource(s) or asset(s) for which rules are applicable, and its definition is very wide - it can be any physical or digital thing that is governed, restricted, or managed by one or more policy rules. It could be an object such as a research output, but it could also be something conceptual such as membership of an organisation (for which a policy applies).
Agent, Party, Actor	These are the natural or legal persons referenced in a policy instance. It may be the rights holder, a data subject, the end user or requestor of access to and use of a resource, the curator of a repository, an organisation providing a service, and so on. It may also be collections of these entities - for example 'researchers'. Some policies or licences are 'agent agnostic' in the sense that they do not reference any agent, and others do, sometimes to the point of explicitly identifying a specific agent instance, e.g. by way of an ORCID for a researcher.

Abbreviations

Term	Expansion
ccREL	Creative Commons Rights Expression Language
CC BY	Creative Commons Attribution licence
DPV	W3C Data Privacy Vocabulary
DUO	Data Use Ontology
eIDAS	Electronic Identification, Authentication and Trust Services (EU Regulation)
EOSC	European Open Science Cloud
GDPR	General Data Protection Regulation
IDS / IDSA	International Data Spaces / International Data Spaces Association
IDSC	IDS information model / Core vocabulary for data space exchanges
IRI	Internationalized Resource Identifier
JSON-LD	JavaScript Object Notation for Linked Data
KB	Knowledge Base
LFS	Licence Facade Service (EOSC EDEN)

ODRL	Open Digital Rights Language (W3C)
PID	Persistent Identifier
RDF	Resource Description Framework
RSL	Really Simple Licensing
SPDX	System Package Data Exchange
TOM	Technical and Organisational Measures
TRL	Technology Readiness Level
W3ID	Permanent Identifier Community namespace

1. Architecture Overview

OpenREL is an ODRL-based rights and policy framework comprising a schema and vocabulary, a policy model, a version-controlled Knowledge Base with reusable instance definitions, encoded policy examples, and reference services supporting policy authoring and evaluation. This specification defines the policy model, the schema and vocabulary and the information model, and describes the associated implementation architecture. OpenREL is also a profile of ODRL 2.2.

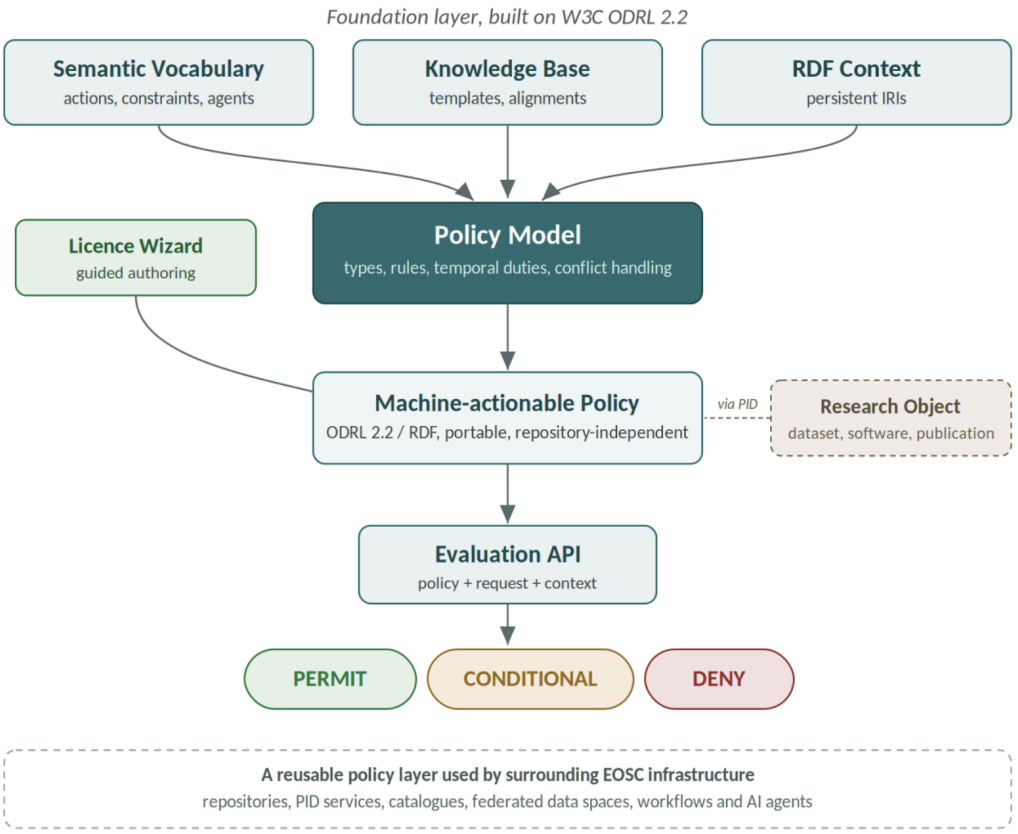


Figure 1. OpenREL at a glance.
A semantic vocabulary, a Knowledge Base and an RDF context feed the policy model. Authored through the Licence Wizard, a machine-actionable policy is evaluated against a request and context by the evaluation API, which returns PERMIT, CONDITIONAL or DENY. A policy is associated with a research object through persistent identifiers. The result is a reusable policy layer that surrounding EOSC infrastructure can build on.

1.1 Components

The OpenREL framework comprises four logical layers:

- **Policy Model**, defining the information model and the evaluation semantics.
- **Vocabulary**, defining reusable classes, properties and controlled terms.
- **Knowledge Base**, providing reusable templates, Named Constraints and canonical policies.
- **Reference Services**, supporting policy authoring, lookup and evaluation.

1.2 Design Goals

OpenREL is shaped by six goals: interoperability, machine-actionability, backward compatibility with ODRL 2.2, extensibility, repository independence, and implementation simplicity. Extensibility is achieved by a conscious design decision to define generalised classes and properties, supported by extensible vocabularies and named instances - this often avoids changing the schema to accommodate new requirements. The chapters that follow trace each design decision back to these goals.

1.3 What OpenREL Is, and Is Not

OpenREL is a common, machine-actionable policy layer. It is not a repository, a licence registry, an identity provider, an access control system or a legal reasoning engine. It provides a shared representation of rights and policies that these systems may use. OpenREL evaluates policy conditions; it does not authenticate identities, execute obligations or enforce access decisions. Those responsibilities remain with the surrounding infrastructure.

1.4 Repository Independence And Portability

OpenREL is intentionally repository-independent. Policies are independent digital resources that may be associated with datasets, software, publications or other research objects through persistent identifiers and standard metadata relationships. Repositories remain responsible for storing and disseminating research objects, while OpenREL provides a reusable machine-actionable policy layer describing the conditions governing their access and reuse. This lets OpenREL be adopted incrementally by existing repositories without changes to their storage architecture or metadata model.

Policies are also portable. The same OpenREL policy may be referenced by multiple repositories, services or workflows without modification. This separation between research objects and policy definitions promotes consistency, reuse and interoperability across infrastructures.

1.5 A Horizontal Layer for EOSC

OpenREL is intended to operate as a reusable policy layer across the EOSC ecosystem. It may support repositories, PID services, metadata catalogues, federated data spaces, workflow engines, APIs and AI agents through a common machine-actionable representation of rights and policies, rather than through platform-specific integrations.

2. Introduction

2.1 The Rights Expression Problem In Open Science

Open science depends on reuse, and reuse depends on knowing what is allowed. In practice this knowledge is fragmented. A licence such as CC BY 4.0 tells a person a great deal, but a licence label alone does not describe who may access a sensitive dataset, under what identity assurance, for how long, from where, or what must happen before and after access. Those conditions live in repository settings, data management plans, ethics approvals and institutional policies, almost always as prose.

Because licences focus on creator and intellectual property rights, it has become common practice to supplement a licence for sensitive data with separate access conditions kept outside the licence. As long as those conditions stay in prose, every access decision needs a human in the loop, every integration between services re-implements the same logic by hand, and every automated agent that touches research data risks acting outside terms it cannot read. Adjudicating access requests by hand is costly and slow, and it is one of the main bottlenecks in automated research workflows. This becomes acute as EOSC moves toward federated data spaces and toward workflows in which software agents request and process data on a researcher's behalf.

2.2 What Machine-Actionable Means

A machine-readable policy is one a system can parse. A machine-actionable policy is one a system can evaluate against a concrete request and turn into a decision. OpenREL targets the second. A policy is machine-actionable when its actions, constraints, obligations and agents are expressed as typed, identifier-based statements that an evaluator can check, and when the possible outcomes are defined in advance. In OpenREL those outcomes are PERMIT, CONDITIONAL and DENY, and a denial always carries a structured, auditable reason.

OpenREL does not aim to replace legal instruments or to make binding legal determinations. It aims to represent the operational consequences of licences, access rules and processing obligations faithfully enough that a system can interpret and act on them, and to refer a request to a human reviewer when it cannot.

2.3 Who This Serves

The specification serves three groups. Implementers in EOSC and adjacent infrastructures need a reference to build repositories, connectors and policy tooling against. Standards and ontology communities need a clear account of how OpenREL extends ODRL and aligns with related vocabularies. Data stewards, policy professionals and reviewers need to understand what OpenREL can express and how policies are authored. The chapters that follow move from design rationale to the model, the vocabulary, the Knowledge Base, the authoring tools, worked examples, the surrounding landscape and the adoption path.

3. Design principles

3.1 Machine-Actionable First

Every OpenREL policy is designed to be evaluated by a machine. Human-readable texts and legal code are not generated by OpenREL. The policy carries properties that link to them, so that the three views of a policy can be reached from one record.

3.2 Backward Compatibility With ODRL 2.2

OpenREL is a profile that builds on ODRL 2.2 rather than a replacement for it. Its policy types are specialisations of odrl:Policy, and it reuses ODRL constructs for permissions, prohibitions, duties, constraints and conflict handling wherever they exist. A consumer that understands ODRL can process the ODRL layer of an OpenREL policy.

3.3 Composability

Real policies are rarely a single licence. A resource can carry a licence, access rules, and a processing rules of obligations at once. OpenREL treats these as parts that can be combined into one evaluable policy rather than forcing them into a single flat statement whose intent a machine would have to reconstruct.

3.4 Legal Neutrality

OpenREL expresses the operational effect of rules without asserting a jurisdiction-specific legal interpretation. It records that a policy is intended to represent a licence with legal significance and that an access rule is operational, and it keeps the two distinguishable, but it leaves legal interpretation to the appropriate authority. Whether a given licence instrument is in fact enforceable depends on jurisdiction, validity, the parties and the actual circumstances.

3.5 Multiple Representations Of The Same Policy

The same policy can be presented as natural language for researchers and data stewards, as a structured permission, prohibition and obligation view for legal and policy professionals, and as RDF for machines. OpenREL does not generate one of these forms from another. The policy carries properties that point to where its texts and encodings are held, so that all three can be found from one policy record.

3.6 Profile-Based Extension, Open And Extensible Vocabulary

The vocabulary is not fixed. It is maintained in a Knowledge Base and grows as new scenarios are encoded or community requests are implemented. New actions, constraints and agent types can be added through changes to the base OpenREL profile (adding classes, properties, or concepts), or by creating community profiles without breaking existing policies. OpenREL introduces new terms only where equivalent, interoperable semantics do not already exist in

ODRL or in aligned vocabularies. OpenREL is intentionally extensible: communities may define or encode additional policy types, actions and constraints, provided they remain consistent with the OpenREL information model.

4. From Gap Analysis to Design Decisions

The design decisions in OpenREL follow directly from gaps identified in the EOSC Beyond report D13.3[1]. ODRL 2.2 is a general-purpose policy expression language and a strong foundation, but on its own it does not carry the open-science vocabulary, the typed distinctions, the reusable patterns or the composition and evaluation conventions that open-science policies need. The question was whether to use plain ODRL constructs, such as odrl:Set, odrl:Offer and odrl:Agreement, or to extend ODRL with a focused profile. Moreover, ODRL focuses on creator rights, whereas the research community requires protection of a wider set of rights, as well as access and processing rules not accommodated by the ODRL schema. OpenREL extends ODRL, for four reasons discussed below.

ODRL's own policy subclasses, such as Set, Offer, Agreement, Request and Ticket, is designed to support the lifecycle role a policy plays in a negotiation between parties.. OpenREL's added classes work on a different and complementary axis: they describe the content and origin of a policy, whether it is a licence, an access rule, a processing rule or an aggregation of these. The two axes coexist, so an OpenREL type can still be used within the ODRL negotiation lifecycle. Figure 2 describes the relationship between OpenREL and ODRL classes, and Figure 3 provides examples of negotiation and request processing workflows enabled by these extensions.

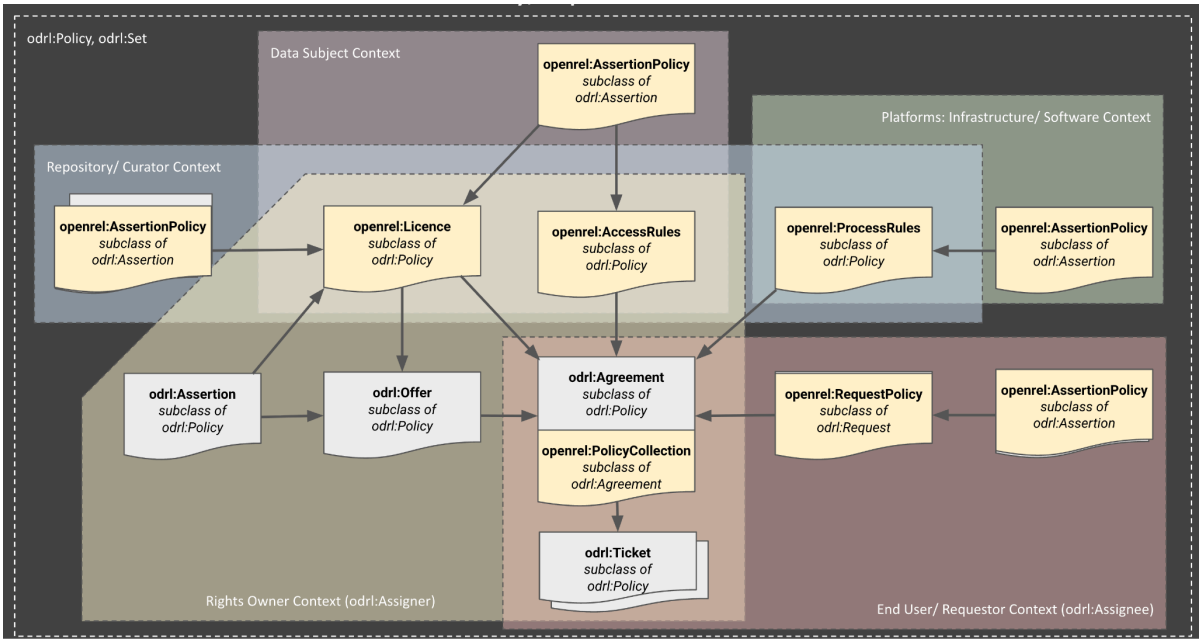


Figure 2. OpenREL extensions of the ODRL Policy class.
The OpenREL subclasses are shown in yellow. It supports three additional contexts that are not present in ODRL: the context of the data subject, the repository or curator of the resource, and the platform or software that archives, processes, or disseminates the resource. ODRL represents only the context of the rights holder, and that of the requestor in part.

[1] https://doi.org/10.5281/zenodo.17225379

4.1 Immediate Type Recognition

Two ODRL-encoded policies look identical to a machine even when one is a copyright-focused licence and the other is an operational access rule. OpenREL introduces explicit policy subclasses - openrel:Licence, openrel:AccessRules and openrel:ProcessRules - so that a system recognises the kind of policy it is handling from the class alone, without inferring intent from the content. This foundational distinction OpenREL adds allows separation of the effects of changes in the way a resource (ODRL Asset) is managed. As an example: an openrel:Licence will be unaffected by moving a resource from one repository to another, since the rights of the creator are unaffected by such a change. The openrel:AccessRules are much more likely to change, since the repository or curator determines the level of protection afforded to data subjects and how that is achieved, and/ or the extent to which ethics clearance is required prior to use of sensitive data, together with many other context-dependent policy provisions that may be in place.

4.2 Representing Open-Ended Licences In OpenREL

Common licences such as CC BY, CC BY-NC and the MIT and GNU software licences are open-ended and non-negotiable: they do not name a specific asset, assignee or assigner, and one cannot enter into negotiation to change the terms of the licence. A licence is typically attached to a resource indirectly, by pasting its text into source code or by naming it in dataset metadata. ODRL's negotiated subclasses, odrl:Offer and odrl:Agreement, are typically used with the asset and the parties specified, so an open-ended licence is not a natural fit for those classes. It can be expressed with a generic odrl:Set, which is capable of carrying rules over an asset, but nothing in an odrl:Set identifies it as a licence. OpenREL therefore introduces openrel:Licence, a non-negotiable policy that stands in for an offer that will not be modified: if a request does not match the terms of the licence, no agreement is possible. Because a licence also carries legal significance, marking it explicitly makes the domain role of the policy machine-recognisable, rather than something to be inferred from external context or from the content of the policy, and lets a system distinguish a licence-type policy from an operational access or processing policy when it weighs the two or records why a decision was reached. These dedicated subclasses fit more naturally into typical use cases encountered in the research landscape, as shown in Figure 3.

ODRL, as shown on the left-hand side of Figure 3, is well suited to a negotiated access - deciding which permissions in a request for access to a resource will be granted by rights holder, with or without restrictions and duties, and making an offer for such access. If the offer is accepted, an agreement can be formulated, and a ticket may be needed to define specifics of such access (time limits, for example). OpenREL allows variations on this workflow. Routinely adding access rules to the licence for a resource (middle workflow) is one such example. This may be required when an embargo applies: it does not alter the licence, but it determines when access can be granted. The Access Rules may express this requirement in parametric terms (e.g. ‘embargoed for 6 months after publication’), while a ticket contains an explicit date based on the publication date of the resource.

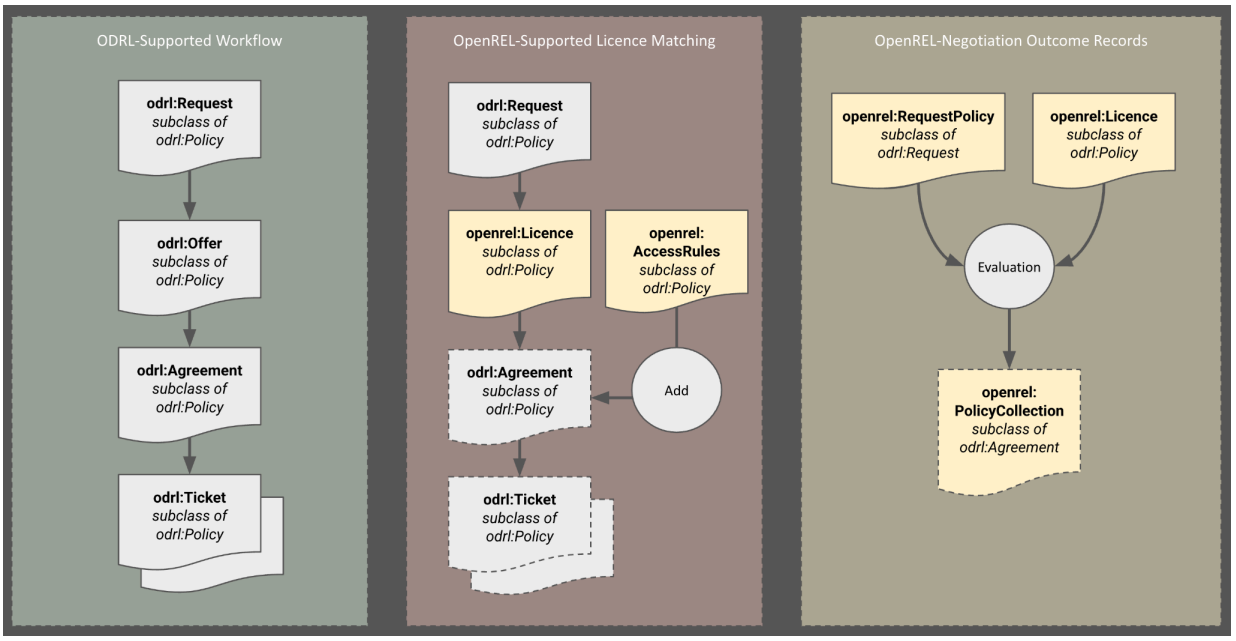


Figure 3. OpenREL-Supported Workflows.
OpenREL extensions support workflows encountered in the research landscape more directly. OpenREL Policy subclasses are shown in yellow.

Records of access requests to resources that have specific constraints on use needs to be kept in the research context, and that is also accommodated by OpenREL. ODRL makes provision for recording agreement, but makes no such provision for cases where access was not granted, or for recording the reasons why. Both of these cases (shown in the right-hand workflow in Figure 3) need to be recorded by repositories for legal, ethical, and transparency reasons.

4.3 Temporal Obligations Before And After An Action

Many open-science obligations are ordered in time. Consent must be obtained before access. A deletion or a log entry must happen after access. ODRL implicitly treats a duty attached to a permission as a pre-condition of that permission, and also supports policy-level obligations and consequences. It does not define a normative way to state that a requirement falls due after the permitted action has been exercised. OpenREL defines openrel:PreDuty and openrel:PostDuty as specialised ODRL Duty classes, which build on the underlying ODRL duty constructs rather than replacing them. An unsatisfied PreDuty MAY lead to CONDITIONAL if it remains satisfiable; an unsatisfiable mandatory condition results in DENY. PostDuty non-fulfilment must be assessed against its due time/conditions; do not treat every not-yet-fulfilled PostDuty as an immediate violation.

4.4 Aggregation Of Rules From Several Sources

The aggregation case, drawn from repository practice at DANS, combines a Creative Commons licence, institutional access conditions and a data processing obligation on a single object. ODRL does not define a standard, interoperable composition construct for holding these together as one evaluable unit. OpenREL introduces openrel:PolicyCollection, a

container that aggregates rules from multiple sources into a single, independently evaluable structure in which each section carries its own rules, constraints and agents.

4.5 OpenREL Positioning in the Landscape

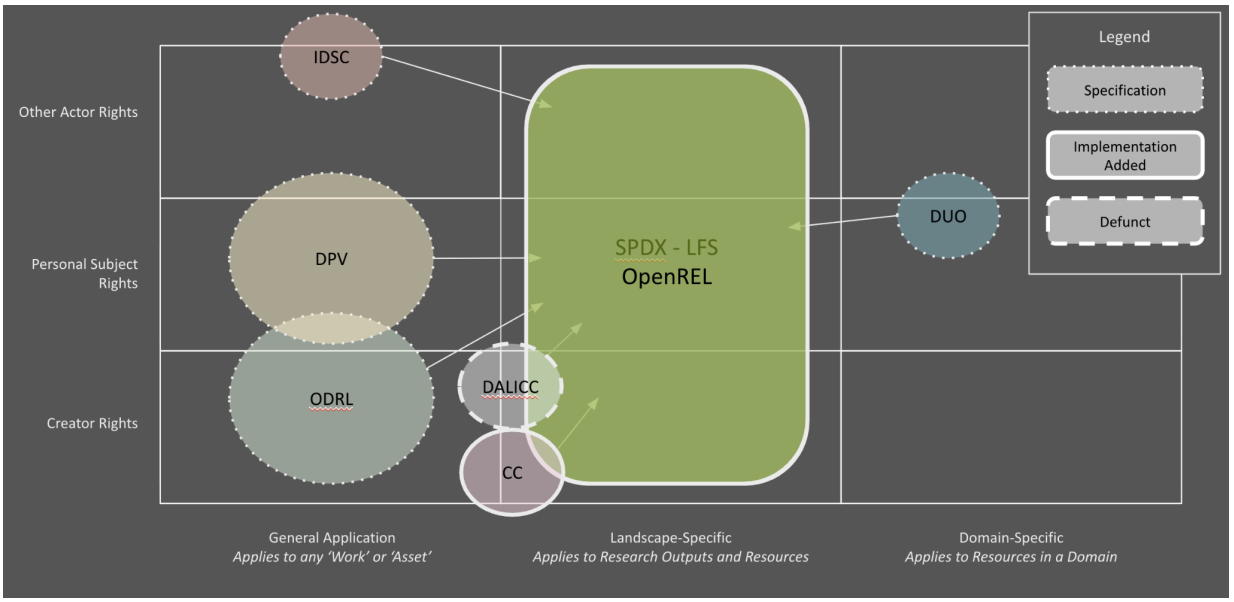


Figure 4. OpenREL Positioning.
Existing resources cover parts of the landscape: ODRL as a general policy expression language, DPV and DUO for the rights of individuals as data subjects, IDSC for data space exchanges, and SPDX, CC and the former DALICC work for licence metadata. OpenREL fills the gaps between them and provides the missing encoding for the rights of all actors.

The use of Licences to define aspects such as copyright, access conditions (if any), and obligations of and limitations on the user of a work is a common feature not only of the Research Data Management landscape, but generally applies to creative work or outputs that are produced for reuse and dissemination. Licensing of creative works in the broader sense was the original driver for the development of ODRL (Open Digital Rights Language) - a W3C standard that defines the concept of a Policy: a container for the Rules (Permissions, Duties, and Prohibitions,) that the rights holder confers on those who want to use the work (Asset). ODRL is widely adopted in many industries and is actively maintained by the W3C. ODRL has some shortcomings, but by far and away the most significant one for practitioners and users involved in the research landscape is that it ***focuses on creator and intellectual property rights***, and is not easily used for the other rights (subject rights in particular) that may apply to the use of a work. More broadly, non-IP considerations such as ethical norms are also out of scope for ODRL.It has, as a result, become common practice to supplement a 'licence' for sensitive data with additional 'access conditions'.

Encoding these considerations can be addressed in part by the DPV (Data Privacy Vocabulary) specification - also a W3C standard - that aligns strongly with GDPR, but it cannot be used to describe the scope and nature of other sensitive data categories. DPV provides a conceptual framework for doing so, and OpenREL adds examples of these required extensions.

Two additional vocabularies are specifically focused on encoding what is generally considered to be a 'licence': CC (Creative Commons) has defined a number of terms that are required to encode Creative Commons Licences, and DALICC (Data License Clearance Center), a now-concluded project, developed a supporting vocabulary and encoded upwards of 350 common licences using a combination of ODRL, CC, and their own vocabulary. The DALICC resources are no longer maintained, even though the project outputs remain available.

There are a number of additional vocabularies that are applicable to the context, but for brevity we highlight only two others: DUO (Data Use Ontology), which is specifically developed to define restrictions on data access in the bioinformatics and health domain, and IDSC (International Data Spaces Core vocabulary), which encodes the processes and protocols for trusted access and exchange of data resources between Data Space nodes. SPDX is maintained by the Linux Foundation, and is a primary registry of software licences with some common data licences (e.g. CC BY 4.0, etc.) included. Anyone can submit a pull request to register a new licence, but SPDX is

- unlikely to register custom or domain-specific licences, for example one used by DANS in the Netherlands for sensitive data - this because the scope is not global,
- certainly will not register policies and licences that represent individual instances with designated parties and assets included.

The EOSC EDEN project is in process of developing the License Facade Service (LFS) that serves as an extension of SPDX for custom licenses and policies.

With the exception of DALICC (now defunct), CC, LFS, and SPDX, none of the resources described above provides infrastructure and services - they are specifications and not implementations. This means that even if it is possible to encode a policy describing all of the permissions granted to, prohibitions on, and duties of the actors involved, finding and reusing that encoding is far from guaranteed. In the context of EOSC, such infrastructure is clearly necessary to simplify widespread use and adoption of encodings.

5. The OpenREL Policy Model

The policy model is the core of OpenREL. It describes how a policy is structured, how its rules are encoded, how conflicts are handled, how obligations are ordered in time, how a single policy is presented in three ways, and how an evaluation resolves to an outcome.

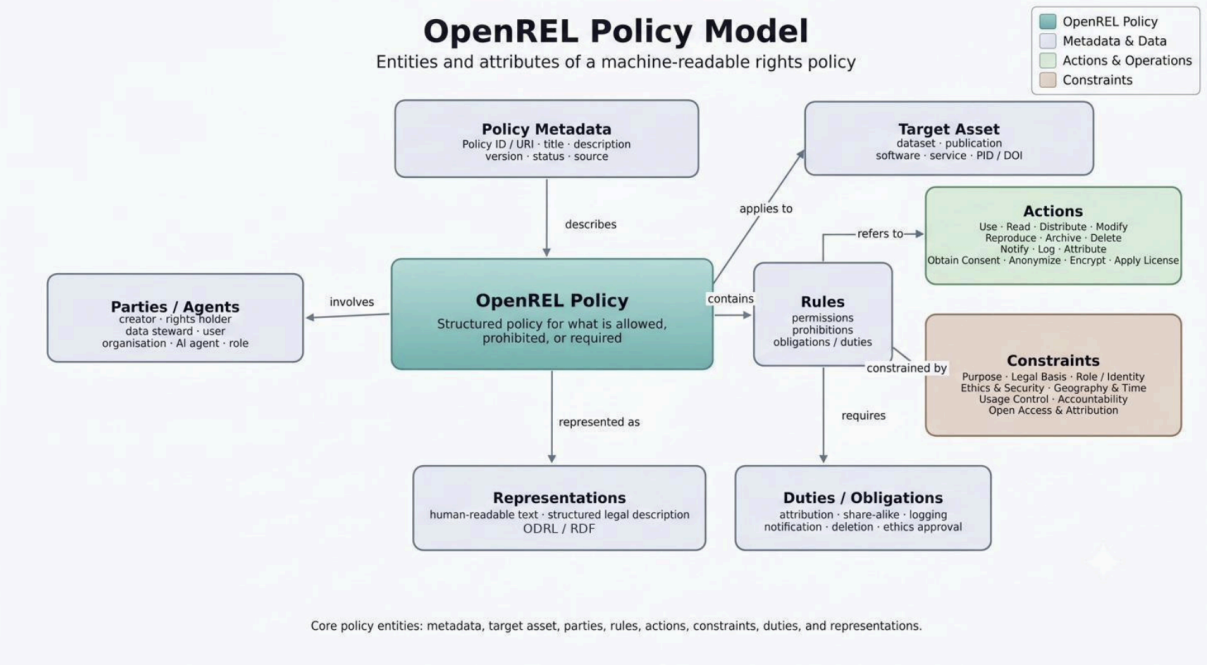


Figure 5. The OpenREL Policy Model.
A policy, carrying its metadata and target asset, is typed and may aggregate sections through a PolicyCollection. It contains permissions, prohibitions and obligations, each referring to vocabulary from the Knowledge Base, can link to its three views, and resolves to PERMIT, CONDITIONAL or DENY.

5.1 Policy Types

OpenREL defines specialised ODRL policy types for licences, access rules, processing rules, policy collections and contextual policies. The latter support recording assertions and requests used in policy evaluation. The principal types are:

- **openrel:Licence**. A rights licence governing use and distribution, for example CC BY-NC. It carries the legal character of the policy.
- **openrel:AccessRules**. Access control rules for a resource: who may access it, when, from where and under what identity assurance.
- **openrel:ProcessRules**. Process-level rules governing how a resource is handled, such as deletion, encryption or consent.
- **openrel:PolicyCollection**. A policy subclass that aggregates rules from several sources into one evaluable policy by referencing at least two constituent policies, each section keeping its own rules, constraints and agents.
- **openrel:AssertionPolicy**. A contextual policy recording claims about parties, assets or other evaluation context through assertions, a subclass of odrl:Assertion.
- **openrel:RequestPolicy**. A contextual policy representing a request and its associated context for evaluation, a subclass of odrl:Request.

5.2 Rule Types And Conflict Handling

A policy[2] contains rules of three kinds. A ***permission*** is an explicit grant of rights to perform an action, always subject to ***constraints*** and possibly carrying ***duties***. A ***prohibition*** is an explicit denial. An ***obligation***, or ***duty***, is a mandatory action attached to a permission, to a set of permissions or to the whole policy.

Across constituent policies, a PolicyCollection requires role-sensitive aggregation of applicable permissions, prohibitions, duties and constraints. The evaluation procedure uses directed comparisons and supporting evidence, and must account for the type, applicability and timing of each rule; a single blanket conjunctive comparison is not sufficient. See the evaluation procedure in Annex B of specification V0.5.2.

5.3 Temporal Obligations

OpenREL distinguishes a preDuty, which must be fulfilled before the permitted action, from a postDuty, which must be fulfilled after it. Obtaining consent is a preDuty; deletion, logging and notification are typical postDuties. A preDuty corresponds to an ODRL duty on a permission, which ODRL defines as a pre-condition of that permission. A postDuty is an OpenREL profile-level construct for requirements that fall due after the action; its serialisation on top of ODRL constructs is confirmed with the profile at vocabulary lock before Version 1.0. This ordering is what lets an evaluator return CONDITIONAL when something must happen first, rather than simply permitting or denying.

5.4 Policies May Have Multiple Representations

OpenREL supports links to different representations of a policy where these are available: a human-readable explanation, a legal or structured representation, and machine-actionable ODRL rules expressed in RDF/Turtle or JSON-LD. The metadata properties openrel:policyText, openrel:policyTextHtml, openrel:legalcode and openrel:encodedRules identify available resources; they do not imply that all representations exist. OpenREL neither synthesises human-readable or legal text from machine-actionable rules nor automatically converts natural-language or legal documents into machine-actionable rules.

5.5 Evaluation Outcomes

Evaluating a request against a policy should result in one of three outcomes. The following table states the condition under which each outcome is returned. It is the evaluation contract an implementation is expected to follow.

Table 5.5.1: Policy Request Evaluation and Policy Comparison Outcomes

Outcome	Required condition

[2] In these descriptions, we use ‘policy’ in the general sense as in ODRL - it includes all and any of the subclasses, such as Licence, AccessRules, ProcessRules, Assertions, Requests, Tickets, Agreements, Offers, and PolicyCollections.

PERMIT	All applicable prohibitions are accounted for and all preconditions, including any preDuties, are satisfied. Obligations may still apply after access, for example required attribution.
CONDITIONAL	No applicable prohibition exists, but one or more preDuties or preconditions remain unsatisfied, for example ethics approval must be submitted before access.
DENY	An applicable prohibition exists, or a mandatory condition cannot be satisfied. A structured, auditable reason is always provided.

Policy comparison and request evaluation are related but distinct operations. The ontology provides openrel:EvaluationDetail for evaluation records and openrel:EvaluationMapping for supporting directed pairwise comparisons, with provenance and rationale. Mapping relationships and semantic assertions are classified separately from the final policy-level outcomes PERMIT, CONDITIONAL and DENY. The evaluator considers applicable rules, the status of prerequisites, duty timing and resolved context before returning a decision. Missing evidence or unresolved parameters require explicit handling and do not, by themselves, establish compatibility. The detailed procedure is specified in Annex B of V0.5.2.

Deciding a single request is one use of evaluation. OpenREL also anticipates comparison between policies, performed by a reasoner. Two comparisons matter in practice: whether two policies are compatible or contain contradictions, which decides whether two assets can be combined and under what terms, and whether one policy satisfies the duties and constraints of another, which is how a request is matched against a licence or an access rule. These comparisons are the basis for combining resources and for recording the outcome of a negotiation. The reasoner is part of the reference services and is currently under development.

Policies can be compared to one another for different reasons, and the results of such comparisons can be recorded explicitly using the PolicyCollection and EvaluationDetail classes. Two or more ‘input’ policies can be referenced in the Policy Collection, and a pairwise comparison can be performed for these policies. The three main comparison situations that apply are:

- Comparing the request for access from a prospective end user (human or machine) against a set of rules that can, in turn, be derived from multiple policies. In a later section, we elaborate the context and assertions that require resolution when such a comparison is made.
- Determining compatibility, for example whether Access Rules or Process Rules imposed by the repository or its software platform are compatible with a Licence that already applies to a specific resource.
- Pre-determining whether a set of input licenses, for example for input datasets to a workflow or input software licences applicable in derived software application, can be combined successfully in a resulting licence[3].

At the root of such comparisons lie deterministic pairwise comparisons of actions and

[3] Creative commons licences that include ‘Share Alike’ and ‘No Derivatives’

constraints, and OpenREL defines the schema and vocabulary for such comparisons. It also provides a non-comprehensive and illustrative registry of such predetermined pairwise comparisons by way of the EvaluationDetail class and its properties. The schema, by design, allows contradictory assertions to be made by different sources - this is likely to happen from time to time.

The following matrix is an illustrative overview only; it is not a substitute for the role-sensitive evaluation procedure and decision rules in Annex B of V0.5.2. In particular, a still-satisfiable precondition differs from an impossible prerequisite, and the timing of post-action duties matters.

Table X provides examples of the typical comparison outcomes when comparing policies. Policy A and Policy B are input policies, and Policy C is an intersection of what the minimum remaining rules should be.

Table 5.5.2 Matrix of Illustrative Comparison Outcomes

Policy A	**Policy B**	**Effective C**	**Aggregate outcome**
Permits read and distribute	Permits read; prohibits distribute	Read only	PERMIT for read; DENY for a request to distribute
Requires authentication	Requires acceptance of licence	Both prerequisites	CONDITIONAL until both are satisfied
Retention ≤ 90 days	Retention ≤ 30 days	Retention ≤ 30 days	PERMIT if all other requirements hold
Requires public distribution	Prohibits public distribution	No valid required distribution	DENY
Requires attribution after use	Requires usage reporting after use	Both PostDuties	PERMIT if other requirements hold; both duties retained

6. The OpenREL Vocabulary

The vocabulary is the ontology layer of OpenREL: the classes, properties and controlled terms that policies refer to. It is expressed in RDF, served under the openrel: namespace from a W3ID persistent identifier, and made available to policies through a JSON-LD context. The vocabulary is open and growing. New terms are added as new scenarios are encoded, so this specification presents it by category and by representative example rather than as a fixed, closed list. Final IRIs are confirmed at vocabulary lock and stated in Version 1.0. The terms available at the time of this draft are archived as a tagged release of the Knowledge Base, identified in the document status table, which is the reference source for the term definitions cited here. Annex A reproduces the actions and constraints of that release as an informative snapshot.

The canonical OpenREL vocabulary namespace is http://www.w3id.org/ns/openrel/0/ (prefix openrel:). This vocabulary namespace is distinct from the ODRL profile identifier, the JSON-LD context document and the versioned Knowledge Base release locations; each is referenced according to its separate function.

6.1 Actions

Actions define operations that may or may not be performed on an asset, and any duties or obligations that are required prior to operations or as a result of it. ODRL defines a substantial number of such Actions natively, and these are left without modification in the OpenREL vocabulary whenever possible. Moreover, other vocabularies, such as DALICC, CCREL, IDSC, and DPV also define numerous Actions. There are many exceptions to the rule of non-duplication, and the main reasons are listed below, see Annex A for a full mapping.

- Extension of the properties of an ODRL Action, often for purposes of mapping: in cases where similar or exact equivalents of an ODRL Action exist on other vocabularies, such as DALICC, CCREL, IDSC, or DPV, these mappings are included in an OpenREL profiled Action with additional semantic mappings between them. The OpenREL Action serves as a semantic pivot for these mapping since the ODRL concept cannot be extended.

- Legacy vocabularies, such as DALICC (which is not actively maintained) are reproduced in the OpenREL vocabulary for convenience. Similarly, existing and often used identifiers, such as the ODRL or CCREL Actions, are also reproduced as stand-alone vocabularies to simplify access via APIs for end-user applications and agents.

6.2 Constraints and Named Constraints

In ODRL, Constraints define the conditions under which a rule applies, and are defined for specific Rules. They are expressed using the ODRL structure of Left Operand, Operator and Right Operand: Since multiple such Constraints can apply to a specific Rule, ODRL defines the Logical Constraint class to allow expression (through logical operators) for compound constraints. ODRL 2.2 allows a Constraint or a LogicalConstraint to carry an identifier through odrl:uid, but it does not provide a ready way for naming and further elaborating Constraints.

This often results in the same Constraint being duplicated in every policy that needs it, and if the encoding is not structurally exactly the same, it complicates comparisons.

An openrel:NamedConstraint is a reusable specialisation of odrl:Constraint, described in Annex A.5 of V0.5.2. Named Constraints provide stable identifiers, labels, definitions and, where appropriate, parameters or references to external endpoints. They can be referenced by rules and used as components of odrl:LogicalConstraint expressions. Logical combinations use the operators supported by ODRL; naming a constraint does not itself define sequencing semantics. The Knowledge Base supports their registration, discovery and reuse.

OpenREL groups its constraint vocabulary into nine categories:

Table 6.2.1: OpenREL Constraint Categories

Category	**What it expresses (representative constraints)**
Purpose	Restricting use to a declared purpose, for example academic research or non-commercial use, classified against the W3C DPV taxonomy.
Legal basis	The GDPR legal basis for processing, for example explicit consent.
Role and identity	The role of the requesting agent, for example researcher or research institution, and identity assurance such as a verified eIDAS identity.
Ethics and security	Documented ethics or IRB approval, IDS security profiles, required technical and organisational measures (TOM), and data minimisation.
Geography and time	Spatial restriction using GeoNames identifiers, a parameterised access duration, and absolute datetime access windows.
Usage control	Access count limits, real-time revocation status, authorised connector identity and data volume limits.
Accountability	Notification endpoints and logging targets such as a clearing house.
Open access and attribution	Open access requirements, attribution, share-alike and public funding identification.
Subject Rights	Broadening of the subject rights of natural persons to include legal persons, and environmental, commercial, and cultural protections.

Several constraints are canonical and parameterised: the value is supplied at policy authoring time. Access duration, for example, takes an ISO 8601 duration parameter, and the Knowledge Base holds predefined profiles for common values. This keeps the vocabulary compact while letting a single constraint serve many concrete policies.

6.3 Agents and Parameters

Agents or parties are sometimes referenced in a policy. OpenREL defines an agent taxonomy for the open-science setting that includes the assigner and assignee of rights, the creator or rights holder, and roles such as data steward, researcher, institution, repository, data provider

and software agent. Each rule carries exactly one action and zero or more constraints; and agents are linked at policy level as assigner or assignee, using the ODRL compact policy form in which properties declared on the policy are shared by all of its rules.

OpenREL also defines openrel:Parameter and a parameter-resolution model (Annex A.7 and Annex C of V0.5.2). A parameter specifies an expected value and can identify a resolution method using the openrel:ParameterBindingSource concept scheme. A Parameter Resolution Service can obtain values from policy declarations, asset metadata, requestor or party context, events, external services and other declared sources. Contextual policies and records—including Assertion Policy, Request Policy, Context Assertion, Resolved Assertion and Resolution Result—allow contextual claims and their resolution evidence to be represented. The model defines these records and their relationships; actual resolution and verification are performed by implementations or supporting services.

Some agents expose a context that is expressed as an Assertion (a special policy type), defining aspects such as the verified affiliation of a researcher with a research institution, or an undertaking not to use derived resources commercially. These are captured as context-defining parameters that are either resolved on demand, or at run-time. OpenREL provides the classes and properties required for encoding of these assertion-like contexts.

Table 6.3.1: OpenREL Parameter Resolution Contexts

Label	Description	Notes
Policy Value	Value explicitly encoded (hardcoded) in a policy.	Example: 'P30D'^^xsd:duration - describing an embargo period of 30 days.
Requestor Context	Supplied in the current access or usage request, reflecting the immediate context of the requesting Party.	Ideally, the context SHOULD be captured in an Assertion Policy associated with the requesting Party, or in a Request Policy. These are values obtained from the requesting Party, the requested action, and/ or current device. Examples: current location, IP Address, intended purpose and use.
Subject Context	Required by a data subject, reflecting assertions in respect of access and use.	Ideally, the context SHOULD be captured in an Assertion Policy or Access Rules Policy associated with the data subject. Examples: inventory of data elements that can be shared.
Repository Context	Required or Supplied by repository or curator.	Ideally, the context SHOULD be captured in an Assertion Policy or Process Rules Policy associated with the repository. Examples: repository IRI, end user identity must be supplied, valid token for use must be supplied, licence conditions must be acknowledged.
Platform Context	Obtained from the software platform on which the asset is hosted, disseminated, or processed.	Examples: whitelist of allowed IPs, robots.txt, rate and volume limits .
Runtime Context	Obtained from the evaluator's current environment.	Examples: current time, location, network.

Policy Usage Value	Obtained from the history or state of this policy's use - may involve a log or clearinghouse call.	Examples: Number of access events, volume of downloads, timestamp of first access.
Event value	Obtained from a named event.	Examples: a workflow state such as 'Published' or 'Approved'.
Asset Metadata Value	Read from metadata about the governed asset.	Examples: Asset PID, governed asset publication date, classification, associated licence, Asset Collection.
Party Context	Characteristics of the requesting Party, reflecting a persistent context of the requesting Party.	Ideally, the context SHOULD be captured in an Assertion Policy associated with the requesting Party, or in a Request Policy. Examples: institutional affiliation, verified identity, confirmation of research role, assertion on non-commercial application.
External Reference Value	Dereferenced from an external service or resource.	Examples: exchange rate for cost calculation in an Agreement Policy, status of an endpoint.
Contract State Value	Obtained from the state of an agreement or workflow.	Examples: date of contract signature, payment status.
Supplied Parameter Value	Explicitly supplied by the end user when instantiating a template or parameterised Policy.	These values are requested from an end user (human or system) if it cannot be resolved by one of the pre-defined methods. For example, it may not be possible to determine the institutional affiliation of the requesting Party from an Assertion or Request, in which case it has to be supplied directly.
Automated Test Result	This value results from a test or verification process via a web-based endpoint or API, using deterministic methods.	Examples: Confirming identity and affiliation via AAI, validating a Ticket, invoking policy comparisons.

7. The Knowledge Base

The Knowledge Base (KB) is the implementation layer that holds the vocabulary and the canonical policies in a form that tools and services can consume. It is modular and version-controlled. Definitions are edited in a structured source and published as JSON files through a Google Sheets to GitHub pipeline, so every change to an action, constraint or policy is tracked.

7.1 Policy Status and Scope, Templates, and Instances

The Knowledge Base distinguishes a Policy Template from a Policy Instance. A template is a reusable, possibly parameterised policy pattern. An instance is a template with concrete values supplied, or that defines the means to resolve open Parameters at run-time. possibly bound to a specific asset and specific parties. The Knowledge Base includes a set of canonical templates that act as ready-to-use building blocks for common EOSC scenarios, from a plain CC BY licence to managed access and processing policies.

Policies can have a state, as indicated in the table below. The state (scope) property describes three interrelated dimensions, if needed: the scope of the policy (Local/ private, custom, or global)'; the publication state (draft, review, or active), and whether it is a template or not. For templates, parameters are indicated by placeholders, and for others, parameters that may be present have been resolved.

Table 7.1.1: ODRL Parameter Resolution Contexts

Label	**Description**
Draft Policy	A policy that is still in draft status. Private by default.
Policy in Review	A policy that is being reviewed. Private by default.
Policy Template	A policy that is used as a template, with placeholders for parameters.
Policy is Active	A policy that is active. Can be private, custom, or global. If parameters are present, they are fully resolved.
Policy is Active/ Private	A policy that is active but applicable in a specific private context only. For example, assigner, assignee, and asset have all been explicitly defined. Example: Data Use Agreement for a specific data set for a specific resource.
Policy is Active/ Custom	A policy that is active but applicable in a specific custom context only. For example, assigner and asset collection may have been explicitly defined - typically for a repository or collection. Example: Repository Sensitive Data Licence.
Policy is Active/ Global	A policy that is active and applicable globally. Typically, assigner, assignee, and asset/ asset collection are not defined. Example: CC BY 4.0 Licence.

Registries of policies contain instances of all of these states, but the extent which they are visible and/or federated is determined by the state. A policy can have multiple state indicators, for example a template can be a draft, and a global policy can be a template.

7.2 Imports and Alignments

OpenREL does not rebuild what others maintain. It caches and links to established vocabularies so that a policy can refer to them directly, and so that the accompanying API can offer a comprehensive service.. Creative Commons and examples of other common licences have been transformed from legacy DALICC encodings to OpenREL-based templates for general use. . Privacy purposes, roles and legal bases align with the W3C DPV. Geographic identifiers come from GeoNames, downloaded once at build time and mapped to country names at authoring time, so evaluation is a local comparison rather than a live network call. Alignment with SPDX for licence metadata is in progress.

7.3 Lookup and Reasoning

A prototype API for the Knowledge Base is in development, testing and refinement. It allows tools to resolve identifiers, retrieve definitions, and fetch canonical templates. A graph database that will support richer reasoning over the vocabulary and the policy corpus is under development and is not yet complete, although example deterministic comparisons between pairs of actions and constraints are already available. Where this specification refers to reasoning support, it refers to a planned capability.

7.4 Independence and Governance

Policies remain valid independently of the Knowledge Base. The Knowledge Base exists to promote reuse, discoverability and consistent encoding, not to determine policy validity: a well-formed OpenREL policy is valid whether or not its terms are also registered in a Knowledge Base.

The OpenREL vocabulary is version-controlled. New terms are reviewed before publication and assigned stable persistent identifiers. Policies remain interpretable across versions through these stable persistent identifiers. The full governance model, including roles and the change process, is documented in Deliverable D14.3.

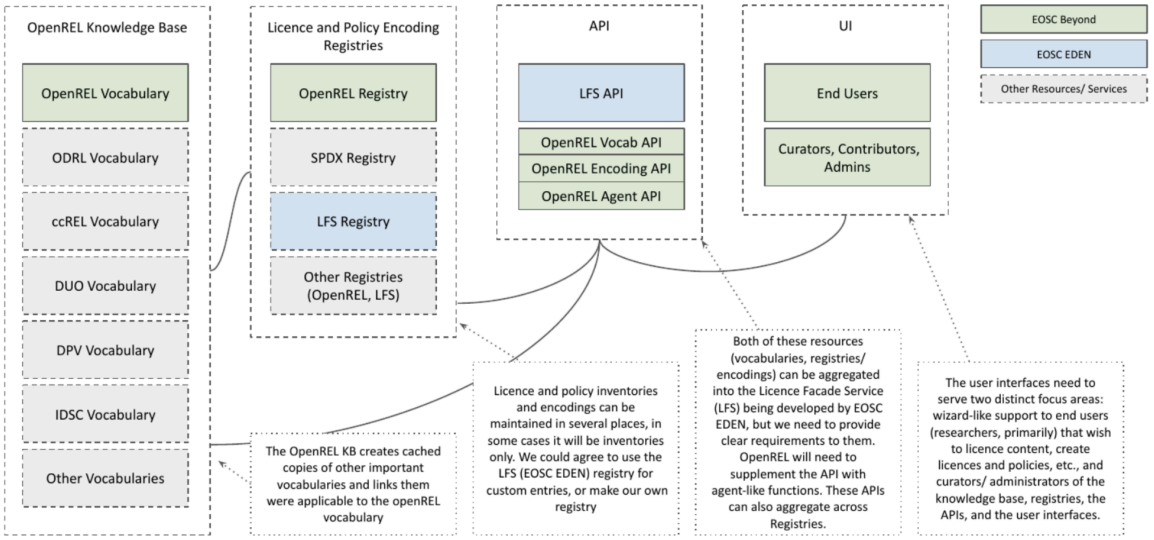


Figure 6. The Knowledge Base. In its intended infrastructure, the OpenREL vocabulary caches and links related vocabularies (ODRL, ccREL, DUO, DPV, IDSC). Encoded licences and policies live in registries, including an OpenREL registry, the SPDX registry and the Licence Facade Service registry developed by EOSC EDEN. APIs expose the vocabulary and encodings to user interfaces for end users and for curators.

8. Authoring and Policy Creation

A machine-actionable policy is only useful if the people who own the rules can produce one. OpenREL treats authoring as a first-class concern. The flagship implemented capability is the Licence Wizard, which lets a data steward compose a policy without writing any code or knowing the vocabulary.

8.1 The Licence Wizard

The Wizard offers two modes. Simple mode asks plain-language questions and assembles the policy from the answers. Advanced mode presents a step-by-step form that exposes the underlying actions, constraints and agents for finer control. In both modes the output is a machine-readable ODRL JSON-LD policy that draws on Knowledge Base vocabulary, so authored policies are consistent with the canonical terms by construction.

The Wizard includes duplicate detection through structural fingerprinting: before saving, it checks whether an equivalent policy already exists, so the registry does not accumulate near-identical entries. This keeps the growing policy corpus clean and reusable.

8.2 What Exists and What is Planned

The Licence Wizard exists, in both modes, with duplicate detection. Validation services, programmatic API access for authoring, and further authoring channels are on the roadmap and are presented here as planned rather than delivered. The boundary between the two is kept explicit throughout OpenREL so that adopters can plan around what is available today.

9. Examples

Three examples show OpenREL in use, one per policy type that a reader is most likely to meet: a licence, a process rule and a composite policy. Each gives the plain-language intent, the metadata summary and the encoding. Examples 9.1 and 9.3 show the rules in Turtle (RDF), excerpted from the policy files of the OpenREL Knowledge Base, with the descriptive metadata left out. Example 9.2 is an illustrative JSON-LD sketch that will be aligned with the Knowledge Base encoding. Identifier placeholders in square brackets are supplied at instantiation. The final namespace and term IRIs are confirmed at vocabulary lock before Version 1.0. The JSON-LD sketch in 9.2 begins with an @context referencing the OpenREL JSON-LD context under the persistent namespace and declares the OpenREL profile through odrl:profile; the placeholder [OPENREL_PROFILE_IRI] stands for the profile identifier, which is not yet fixed and will be confirmed at profile and vocabulary lock before Version 1.0. It also advertises, through openrel:possibleOutcome, only the outcomes it can actually produce.

9.1 A Licence: CC BY 4.0

This is an illustrative, simplified excerpt of the OpenREL encoding of the Creative Commons Attribution 4.0 International licence, as held in the OpenREL Knowledge Base. It shows the rules only; descriptive metadata is given in the summary table. It does not encode every legal condition of the licence, and the authoritative legal instrument remains the Creative Commons licence itself. Other permissions of the encoding are omitted. Displaying, presenting and reproducing are permitted. Distributing, modifying and deriving are permitted subject to attribution and notice duties. For an evaluator, a compliant request receives PERMIT together with the attribution duty. CC BY 4.0 is a widely used open-access licence in the research and EOSC ecosystem, and a natural starting point for the licence vocabulary.

Table 9.1.1: OpenREL Encoding Example - CC BY 4.0

Policy ID	CC-BY-4.0
Type	openrel:Licence
Source	https://creativecommons.org/licenses/by/4.0/
SPDX	CC-BY-4.0

The first example indicates how a Licence is encoded with SPDX-compliant licence-level metadata (RDF TTL example).

```
<CC-BY-4.0> a openrel:Licence ;
    # Attribution
    prov:wasDerivedFrom dalicclib:CC-BY-4.0 ;
    prov:wasAttributedTo "Giray Havur" ;

    # Alternate identifiers
    dct:relation spdxlicense:CC-BY-4.0 ;
    dct:relation lfslib:CC-BY-4.0 ;
```

```
    # Citation Information
    dct:title "Creative Commons Attribution 4.0 International"@en ;
    dct:alternative
        "Attribution 4.0",
        "Attribution 4.0 International",
        "CC BY 4.0",
        "CC BY 4.0 International",
        "CC-BY 4.0",
        "CC-BY-4.0",
        "CC-BY-4.0 International",
        "Creative Commons BY 4.0",
        "Creative Commons BY 4.0 International" ;
    dct:description "Anyone can read, download, share, and adapt for any
purpose, provided the original author is credited."@en ;
    dct:publisher "Creative Commons" ;
    dct:created "2026-07-12T00:00:00+00:00" ;
    dct:modified "2026-08-03T00:00:00+00:00" ;
    dct:creator "Melios Katsamakis - OpenAIRE" ;
    dct:contributor <https://orcid.org/0000-0002-0255-5101> ;
    dct:subject "Public Domain" ;
    openrel:accessType openrel:openAccess ;
    dct:source <https://creativecommons.org/licenses/by/4.0/> ;
    dct:version "4.0" ;
```

The metadata can be extended to include information about legal and human-readable representations, as well as coverage and warranty-related metadata. In the case of a Creative Commons licence encoding, aspects such as the warranty description corresponds to information provided by Creative Commons :

```
# Coverage related information
openrel:jurisdiction openrel:worldwide ;
openrel:validityType openrel:perpetual ;
openrel:policyStatus openrel:typeGlobalActive ;
# Disclaimers
openrel:liabilityLimitation "To the extent possible, in no event will the
Licensor be liable to You on any legal theory (including, without limitation,
negligence) or otherwise for any direct, special, indirect, incidental,
consequential, punitive, exemplary, or other losses, costs, expenses, or
damages arising out of this Public License or use of the Licensed Material,
even if the Licensor has been advised of the possibility of such losses, costs,
expenses, or damages. Where a limitation of liability is not allowed in full or
in part, this limitation may not apply to You." ;
openrel:warrantyNotice "Unless otherwise separately undertaken by the Licensor,
to the extent possible, the Licensor offers the Licensed Material as-is and
as-available, and makes no representations or warranties of any kind concerning
the Licensed Material, whether express, implied, statutory, or other. This
includes, without limitation, warranties of title, merchantability, fitness for
a particular purpose, non-infringement, absence of latent or other defects,
accuracy, or the presence or absence of errors, whether or not known or
discoverable. Where disclaimers of warranties are not allowed in full or in
part, this disclaimer may not apply to You." ;
# Alternative representations, mappings of SPDX and LFS properties
openrel:legalcode    <https://creativecommons.org/licenses/by/4.0/legalcode> ;
openrel:policyTextHtml <https://spdx.org/licenses/CC-BY-4.0.html> ;
```

```
openrel:policyText
<https://lfs.labs.dansdemo.nl/api/v1/licenses/CC-BY-4.0/original> ;
openrel:encodedRules openrellib:CC-BY-4.0 ;
```

There are four interrelated representation properties to take note of, as shown in Table 9.1.2.

Table 9.2.2: OpenREL Policy Representation Metadata Options

IRI	Label	Description
openrel:legalcode	Legal code	Links to the authoritative legal-text representation of a licence or policy.
openrel:policyTextHtml	Policy text (HTML)	Links to an HTML representation of the policy text.
openrel:policyText	Policy text resource	Links to a retrievable representation of the policy text.
openrel:encodedRules	Encoded rules	Links to a machine-readable encoding of the policy rules.

Encoding of the rules (permissions, prohibitions, and duties) and constraints can be done as follows:

```
#-----------------------------------------------------------------------
# Rules
#-----------------------------------------------------------------------
odrl:permission
   [ a odrl:Permission ;
     odrl:action odrl:use ;
     odrl:duty
        [ a openrel:PostDuty ;
          odrl:action openrel:attribute ]
    ]
.
```

The important point here is that unlike ODRL, OpenREL explicitly indicates that the attribution requirement is an openrel:PostDuty, indicating that the duty need not be evaluated prior to access being given to the Asset.

9.2 A Process Rule: Researcher-only Access with eIDAS

This restricts access to agents that hold a verified researcher role and a valid eIDAS digital identity. A faculty member authenticated through an eIDAS-compliant provider can read the resource; an unverified user or an automated script without credentials cannot. The evaluator checks two attributes from the access token, role and identity assurance, and both must hold simultaneously. This is the standard managed-access pattern for sensitive datasets in EOSC. Both conditions are process requirements on the requestor, so the policy is typed openrel:ProcessRules.

Table 9.2.1: OpenREL Encoding Example - Researcher Verification

Policy ID	openrel:policy.p011
Type	openrel:ProcessRules
Constraints	role: researcher, and identity: verified eIDAS
OpenREL IRI	<OR-BY-AC-VI-4.0>

The encoding below introduces some additional features of OpenREL that are not supported by ODRL in its current release, and that is to indicate parameter placeholders in the policy text. The policy can be regarded as a template for use by anyone requiring such a policy, on the basis that the parameters will then be resolved manually or automatically.

```
<OR-BY-AC-VI-4.0>
    a openrel:ProcessRules ;

    # resolve at run-time from appropriate context
    odrl:assigner openrel:parameterAssignerIRI ;
    odrl:assignee openrel:parameterAssigneeIRI ;
    odrl:target openrel:parameterTargetIRI ;

    dct:title "OpenREL Academic eIDAS Verified Identity"@en ;

    dct:alternative "eIDAS Researcher Verification"@en ;

    dct:description
        """Process rules requiring eIDAS-verified identity
        and verified researcher status. Researcher status
        may be established either through an accepted
        eIDAS assertion that includes researcher status,
        or through independent trusted verification
        combined with eIDAS identity verification."""@en ;

    dct:publisher "OpenREL" ;
    dct:created "2026-07-12T00:00:00+00:00"^^xsd:dateTime ;
    dct:modified "2026-11-04T00:00:00+00:00"^^xsd:dateTime ;
    dct:creator <https://orcid.org/0000-0002-0255-5101> ;

    dct:subject
        "Identity Verification",
        "eIDAS",
        "Researcher" ;

    openrel:accessType openrel:managed ;
    dct:version "0.5" ;

    openrel:jurisdiction openrel:worldwide ;
    openrel:validityType openrel:perpetual ;
    openrel:policyStatus openrel:typeTemplate ;

    # Replace with the published RDF resource IRI.
    # openrel:encodedRules <PUBLISHED_POLICY_RDF_IRI> ;
```

Encoding of the rules (permissions, prohibitions, and duties) and constraints can be done as follows, and introduces an additional feature of OpenREL: a reusable named constraint. It also illustrates the use of an ODRL logical constraint, where both named constraints have to be true for permission to be granted.

```
# ----------------------------------------------------------
# Rules
# ----------------------------------------------------------

   odrl:permission [
      odrl:action odrl:use ;
      odrl:constraint <11-eidas-researcher>
   ] .

# ----------------------------------------------------------
# Logical Constraints
# ----------------------------------------------------------

<11-eidas-researcher>
   a odrl:LogicalConstraint ;
   odrl:or (
      <11-verifiedResearcher>
      openrel:eidasVerifiedResearcher
   ) .

<11-verifiedResearcher>
   a odrl:LogicalConstraint ;
   odrl:and (
      openrel:verifiedResearcher
      openrel:eidasVerified
   ) .
```

The logical constraints illustrate a more complex evaluation of constraints. For permission to use to be granted, a logical constraint has to be satisfied. The logical constraint process is as follows:

- For access to be granted, the requestor must either be a verified researcher, as determined by the constraint <11-verifiedResearcher>, or already have been assessed as an eIDAS verified researcher.
- To determine cases where the researcher is not eIDAS verified (<11-verifiedResearcher>), a workaround is to affirm that the requestor is both a researcher (openrel:verifiedResearcher) and that the identity has been verified using eIDAS (openrel:eidasVerified).

There are three named constraints used in the example, these constraints are encoded as below:

```
openrel:eidasVerified a openrel:NamedConstraint ;
   rdfs:label "eIDAS Verified Identity"@en ;
```

```
    skos:definition "End user or agent has been verified via eIDAS."@en ;
    odrl:leftOperand openrel:contextLimitation ;
    odrl:operator odrl:eq ;
    odrl:rightOperand openrel:context-eidas ;
    openrel:constraintType openrel:context .

openrel:eidasVerifiedResearcher a openrel:NamedConstraint ;
    rdfs:label "eIDAS Verified Researcher"@en ;
    skos:definition "End User has been verified as a Researcher via eIDAS."@en
    ;
    odrl:leftOperand openrel:contextLimitation ;
    odrl:operator odrl:eq ;
    odrl:rightOperand openrel:context-eidas ;
    openrel:constraintType openrel:context .

openrel:verifiedResearcher a openrel:NamedConstraint ;
    rdfs:label "Asserted Researcher"@en ;
    skos:definition "A general assertion or inference that
the end user has the role of a researcher."@en ;
    odrl:leftOperand openrel:agentRole ;
    odrl:operator odrl:eq ;
    odrl:rightOperand openrel:researcher ;
    openrel:constraintType openrel:role ;
    openrel:scenario openrel:academic-only,
       openrel:affirmed-research,
       openrel:confirmed-researcher,
       openrel:confirmed-rpo,
       openrel:research-only .
```

9.3 A Composite Policy: Embargo, then Open Access

This uses a PolicyCollection that references two existing policies instead of repeating their rules: an embargo rule set and the CC BY 4.0 licence of example 9.1. The collection points to its input policies with openrel:hasPolicy and to its own encoded rules with openrel:encodedRules. The embargo rule permits use on or after the publication date, expressed as a comparison on odrl:dateTime against a publication date parameter; the CC BY 4.0 licence then governs that use, with attribution. A staff-only access period before the embargo ends, which an institution may wish to add, is not encoded in this example. Combining separate policies in one evaluable unit while preserving the type and provenance of each is what the PolicyCollection mechanism adds to plain ODRL.

Table 9.3.1: OpenREL Encoding Example - Composite Policy with Embargo

Policy ID	OR-BY-EM-4.0
Type	openrel:PolicyCollection
Constraints	publication date (odrl:dateTime)
OpenREL IRI	<OR-BY-EM-4.0>

The example illustrates an important principle: embargo is an Access Rule, imposed by a repository or curator, and does not alter the terms of a licence that applies to the asset. If the

embargo period has passed, the licence applies as under normal circumstances. The fact that a licence co-exists with the access rules is encoded in an OpenREL Policy Collection:

```
<OR-BY-EM-4.0> a openrel:PolicyCollection ;
  openrel:hasPolicy openrellib:OR-EM-0.4 ;
  openrel:hasPolicy openrellib:CC-BY-4.0 ;
  openrel:encodedRules openrellib:OR-BY-EM-4.0 .

# Input policy 1: openrellib:OR-EM-0.4 (embargo), rules only
openrellib:OR-EM-0.4
  odrl:permission [
    odrl:action openrel:use ;
    odrl:constraint [
      odrl:leftOperand odrl:dateTime ;
      odrl:operator odrl:gteq ;
      odrl:rightOperand openrel:parameterPublicationDate
    ]
  ] .

# Input policy 2: openrellib:CC-BY-4.0, see example 9.1
```

The publication date is normally a stable property of the governed asset. Its value may be obtained from asset metadata when the policy is instantiated or resolved later by an evaluation service; it need not be retrieved afresh for every request. An embargo expiry date is a separate derived value and must account for the applicable embargo duration. The example above is retained pending replacement with the corrected policy encoding.

```
openrel:parameterPublicationDate
    a openrel:Parameter ;
    skos:prefLabel "Asset Publication Date"@en ;
    skos:definition "The publication date of an Asset,
potentially used for embargo calculations."@en ;
    openrel:parameterName "TARGET_PUBLICATION_DATE" ;
    openrel:expectedDatatype xsd:dateTime ;
    openrel:resolutionMethod openrel:AssetMetadataValue .
```

The parameter definition identifies Asset Metadata as the source of the publication date. A resolution service may obtain and bind this value at instantiation or evaluation time, depending on implementation and freshness requirements. This is not, by itself, the embargo end date.

10. OpenREL in the Landscape

This chapter positions OpenREL against the resources a reader is most likely to compare it with, using a defined set of requirements drawn from open-science practice. The comparison concerns what each resource can express out of the box for these requirements. ODRL in particular supplies the underlying machinery, namely constraints, duties and conflict handling, that OpenREL builds on; the matrix reflects domain-specific, out-of-the-box open-science support rather than the theoretical expressiveness of ODRL, which is a general-purpose policy language that can be extended to cover many of these requirements.

Table 10.1. Coverage matrix. Yes, Partial and No indicate out-of-the-box expressiveness for each requirement. Repository metadata refers to the rights fields used by repositories such as Zenodo and DANS.

Requirement	ODRL 2.2	DPV	SPDX	ccREL	DALICC	Repo. metadata	OpenREL
Machine-readable licence terms	Partial	Partial	No	Partial	Yes	No	Yes
Conditional and managed access	Partial	No	No	No	No	No	Yes
Obligations and duties	Yes	No	No	No	Partial	No	Yes
Temporal duties (before/after)	Partial	No	No	No	No	No	Yes
GDPR-aware constraints	Partial	Yes	No	No	No	No	Yes
Geographic constraints	Yes	Yes	No	No	Partial	No	Yes
Composite multi-source policy	Partial	No	No	No	No	No	Yes
Defined evaluation outcome	Implicit	No	No	No	Partial	No	Yes
Legal vs operational typing	No	No	No	No	No	No	Yes

The pattern is consistent with Figure 4. Each existing resource covers part of the landscape well. ODRL provides the general mechanisms but not the domain vocabulary or the typed distinctions. SPDX is a licence registry rather than a rule encoding. Creative Commons and ccREL cover their own licences. The DALICC project produced valuable prior work on licence vocabulary and a reasoner, but its resources are no longer actively maintained. Repository rights metadata is descriptive rather than evaluable. OpenREL is designed to occupy the gap between them and to make the combination evaluable.

11. Conformance

Conformance requirements are defined in the stable OpenREL Specification V0.5.2, rather than restated here. See Annex A.4.6 for evaluation records and provenance, Annex B.8 for evaluation conformance, and Annex C.8 for contextual assertions and parameter resolution. These sections distinguish vocabulary, encoding, evaluation and service responsibilities.

The requirements below are a provisional summary and must be reconciled with the numbered normative statements in Annexes A, B and C before publication. In particular, the

JSON-LD context must not be presented as universally mandatory for RDF/Turtle-only implementations without a corresponding normative requirement.

11.1 Normative Language

The key words MUST, MUST NOT, SHOULD and MAY have the meanings defined by RFC 2119 and RFC 8174 when capitalised. The table below is provisional pending a complete numbered inventory of normative statements in the V0.5.2 specification; the annexed specification remains the authoritative reference.

Table 11.1 High-Level Conformance Requirements

Ref	Requirement
C1	An OpenREL policy MUST use the published OpenREL namespace and the OpenREL policy classes, and MUST declare the OpenREL profile identifier through odrl:profile.
C2	Every OpenREL policy MUST declare one of the OpenREL policy types defined in Section 5.1.
C3	A PolicyCollection MUST retain the type and provenance of each section it aggregates.
C4	An implementation MUST NOT silently ignore unknown terms, and SHOULD surface them for review.
C5	An evaluator MUST return exactly one defined outcome, PERMIT, CONDITIONAL or DENY, together with a structured reason.
C6	An evaluator MUST return CONDITIONAL for an unmet but still satisfiable mandatory precondition, and DENY when an applicable prohibition exists or a mandatory prerequisite cannot be satisfied. The evaluation MUST consider duty timing and the detailed procedure in Annex B.
C7	An implementation SHOULD reference reusable Named Constraints by IRI rather than restating equivalent constraints inline.
C8	An implementation MAY refer an unevaluable request to human review, provided the review resolves to a defined outcome.
C9	Vocabulary-level conformance is out of scope for this draft and is defined in Version 1.0.

11.2 Normative Status and Versioning

Only the conformance requirements in this section are normative. Capabilities described elsewhere as planned are informative and do not create obligations on an implementation.

OpenREL follows a simple versioning policy. Minor versions may introduce new terms without changing the semantics of existing ones. Breaking changes are reserved for major versions. Graduated conformance levels, which will state the capabilities an implementation supports, will be defined in Version 1.0; this draft defines structural conformance only.

12. Adoption Path and Known Gaps

OpenREL is at proof-of-concept maturity, aiming at technology readiness level (TRL) 4 within EOSC Beyond, with TRL 5 where possible. This chapter states the main gap openly and describes how it is intended to close.

12.1 The Repository Integration Gap

There is currently no native automated path from an authored OpenREL policy to a deposit in a repository such as Zenodo, or the other way round. A steward can produce a policy with the Wizard, but linking it to the deposited object still requires a manual step. This is an acknowledged gap in the roadmap.

The interim convention is to relate a policy to the object it governs using a DataCite relatedIdentifier with the relation type isDescribedBy, so the link is at least explicit and resolvable. ODRL 2.2 also defines odrl:hasPolicy as the relationship that links an asset to a policy. The two mechanisms operate at different layers: the DataCite relatedIdentifier is a repository metadata convention, while odrl:hasPolicy is the ODRL-level statement of the same relationship. Neither is automated today. The roadmap components that are intended to close the gap is the Licence Facade Service, developed by EOSC EDEN, which registers policies and licences and can submit them onward to SPDX, if required. PID-based linking, in which a policy is resolved and retrieved from a persistent identifier, is a planned capability rather than a delivered one. Native repository integrations may be implemented through repository plugins, APIs or metadata connectors, so there is no single prescribed path.

12.2 Deployment and Federation

The intended deployment is federated. OpenREL vocabularies and schema definitions are hosted in a single node, but exposed through multiple nodes via their respective APIs. A minimum OpenREL node bundles an instance of an encoded policy registry, an LFS registry instance, the API[4] and the user interfaces. Additional nodes can serve their own communities, nationally, thematically or by discipline, while the LFS layer keeps registries synchronised. This lets the service scale without a single node having to serve every community, and it demonstrates e-infrastructure federation in EOSC. Within EOSC Beyond, options for deploying this suite of proof-of-concept resources on EOSC e-infrastructure are to be discussed with EGI.

[4] This is available as a demonstrator service from the EOSC EDEN project: https://lfs.labs.dansdemo.nl/docs#/

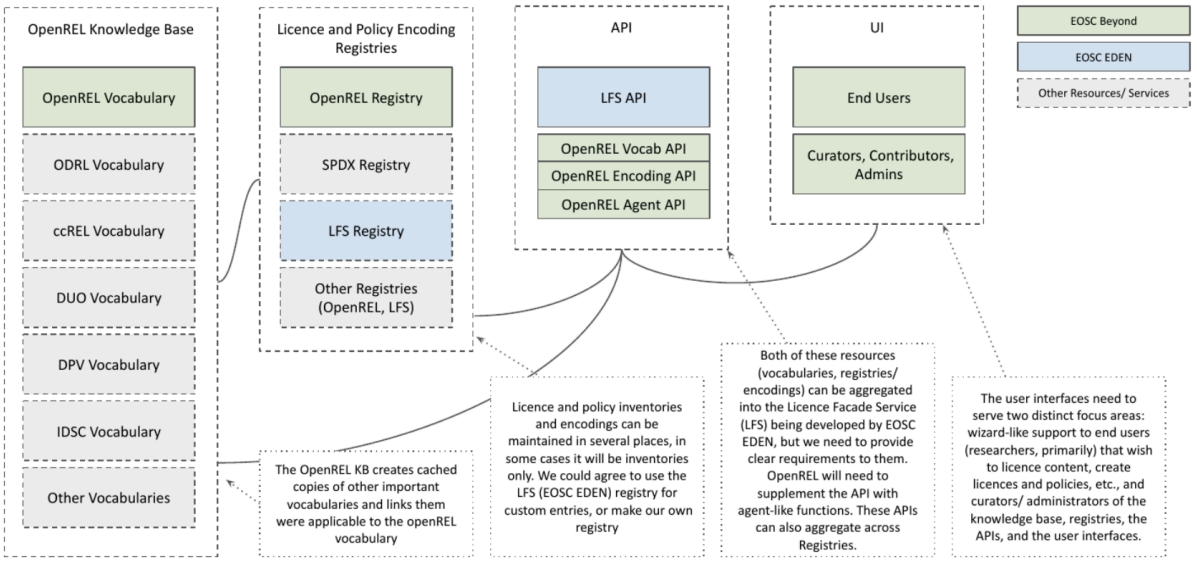


Figure 7. A federated deployment. The OpenREL Knowledge Base and vocabulary API ideally exist once, while multiple nodes host registries, APIs and user interfaces. LFS registries synchronise licences and policies across nodes; the SPDX registry, maintained by the Linux Foundation, acts as an upstream primary registry for licences it accepts.

12.3 Security and Privacy Considerations

OpenREL does not guarantee policy authenticity, identity verification or secure policy distribution. These concerns are addressed by the surrounding infrastructure. In particular, OpenREL allows evaluation of the conditions a policy or policies express, but it relies on the deploying system for authenticating agents, for the integrity and provenance of a policy in transit and at rest, performing evaluation, and for acting on an evaluation outcome.

OpenREL defines vocabulary and structures for representing policy conditions and contextual assertions, which may concern identity, affiliation or other personal information. The vocabulary itself does not authenticate people or perform external verification. Implementations, including evaluation and parameter-resolution services, may process personal data when resolving assertions and requests and must address the applicable data-protection, security, minimisation and accountability obligations. Where conditions refer to personal-data processing, OpenREL can use aligned terms from the W3C Data Privacy Vocabulary (DPV); responsibility for the actual processing remains with the relevant deploying organisations and services.

OpenREL extends beyond the scope of GDPR and DPV, with example encodings of limitations on the use of and classifications of non-personal sensitive data, such as commercially sensitive, environmentally sensitive, culturally sensitive and security-related data. This work is not comprehensive or authoritative at present, but provides a blueprint for this much-needed capability.

12.4 Future Work

Several directions are already identified for the roadmap to Version 1.0 and D14.3.

- **IDSA Core alignment.** Given the direction of EOSC toward data spaces, aligning OpenREL with the IDS information model is the most pressing of the alignment tasks. This task has already started, with selected mappings between IDSA Core concepts and classes and OpenREL included in the Knowledge Base. This is already partly completed in version 0.5.2.
- **DUO alignment.** Aligning with the Data Use Ontology and its successors to cover consent-based data use conditions in health and human-subject research. This is already partly completed in version 0.5.2.
- **RSL mapping.** A mapping layer to Really Simple Licensing, which handles enforcement plumbing while OpenREL provides policy semantics. Because RSL is XML-based without an RDF form, this requires manual vocabulary mapping.
- **Machine validation profiles.** Structural validation profiles, for example using SHACL, so that policies can be checked automatically against the OpenREL information model.
- **Graph database completion.** Completing the graph database of mappings to support richer reasoning over the vocabulary and the policy corpus. This is already partly completed in version 0.5.2.
- **Pilot adoption.** Testing the model against real repository and Data Space scenarios and folding the feedback into the normative specification.

13. Conclusion

OpenREL turns the rights that govern research objects from prose that only people can read into structured statements that systems can interpret and act on. It does this by extending a proven standard rather than replacing it: ODRL supplies the mechanisms, and OpenREL adds the typed distinctions, the temporal obligations, the composition mechanism and the open-science vocabulary that ODRL alone does not carry. Further distinctive contributions are pairwise comparison of actions and constraints between policies, by-reference policy collections that combine existing policy instances, and customisable concept schemes. Around the model sit a version-controlled Knowledge Base, authoring tooling led by the Licence Wizard, and a federated deployment path.

This is a first public draft, published to be built against and argued with. Its gaps, above all the repository integration gap and the still-evolving vocabulary, are stated openly. The normative Version 1.0 will follow with EOSC Beyond Deliverable D14.3, incorporating community feedback and pilot experience. Comments on this draft are welcome and will shape that version.

In the longer view, OpenREL aims to provide for research rights what shared metadata standards provided for research description: a common, interoperable representation that lets heterogeneous systems exchange, interpret and act on policy information consistently across the research landscape.

References

The following are the primary standards and resources OpenREL builds on. Additional citation records and any assigned DOIs are included in the published Zenodo record.

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